

Question one

- (a) Search is a fundamental problem-solving technique in AI that involves exploring a state space to find a path from an initial state to a global state.

Major differences:

<u>Aspect</u>	<u>Uniformed Search (Blind Search)</u>	<u>Informed Search (Heuristic Search)</u>
Knowledge	No additional information about states	Uses domain-specific heuristic knowledge
Guidance	Explores blindly without direction	Uses heuristic function to estimate cost to goal
Efficiency	Less efficient, explores many unnecessary states	More efficient, focuses on promising paths

(c) (i) <u>Concept</u>	<u>Definition</u>
Valid (Tautology)	True under all interpretations
Satisfiable	True under at least one interpretation
Unsatisfiable (Contradiction)	False under all interpretations
Falsifiable	False under at least one interpretation

Question two

- (a) PEAS and PAGE for DNA/Genomic Sequencing System

PEAS Framework (Performance, Environment, Actuators, Sensors)

<u>Component</u>	<u>Description</u>
Performance Measure	• Accuracy of DNA matching ($\geq 99.99\%$) • Speed of identification (< 48 hours) • Number of successful identifications • False positive/negative rates • Cost per sample processed
Environment	• Forensic laboratory setting • Partially observable (DNA sequences) • Deterministic (DNA doesn't change) • Static (samples remain unchanged) • Discrete (distinct genetic markers) • Known environment (controlled lab)
Actuators	• DNA sequencer control • Database query interface • Alert/notification system • Report generator • Sample tracking system
Sensors	• DNA sequencing machine • Barcode scanner for samples • Temperature sensors (sample preservation) • Mass spectrometer • PCR thermocycler sensors

PAGE Framework (Percepts, Actions, Goals, Environment)

Component	Description
Percepts	• Raw DNA sequence data • Sample quality metrics • Contamination indicators • Database match scores • Chain of custody data
Actions	• Sequence DNA samples • Compare against reference database • Flag potential matches • Generate identification reports • Request re-sequencing if low quality
Goals	• Identify each victim correctly • Minimize identification time • Maintain chain of custody • Maximize sensitivity/specificity • Produce admissible forensic evidence
Environment	• Laboratory with DNA databases • Controlled temperature conditions • Automated sequencing equipment